

Culturally-Aware Visual Captioning for Indigenous Languages of the Americas

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We participate in the AmericasNLP 2026 Shared Task on Cultural Image Captioning for Indigenous Languages (CICIL), which asks systems to generate captions for culturally situated images directly in the target indigenous language — Guaraní, Yucatec Maya, Bribri, or Wixárika. These are endangered languages spoken by communities actively pursuing revitalization, and generating culturally grounded educational materials at scale is a concrete, unmet need. The shared task winner identified the vision stage as the key bottleneck — generic VLMs discard cultural information before translation begins — but did not address it. Our goal is to fix this with a culturally-aware vision encoder and study whether doing so actually moves the needle. Three research questions drive the paper: (RQ1) Does culturally-aware vision encoding improve end-to-end caption quality over generic VLM baselines? (RQ2) Does the benefit vary by language resource level — helping more for data-starved Wixárika and Bribri than for relatively data-rich Guaraní? (RQ3) Which cultural categories are hardest for current VLMs to ground correctly, and does this correlate with caption quality?

We use three existing datasets. The primary source is the AmericasNLP 2026 CICIL dataset: culturally situated images paired with indigenous language captions, with 50 development examples per language and Spanish reference captions in the pilot set, which serve as our Stage 1 fine-tuning signal. For Stage 2 retrieval we draw on Spanish↔Indigenous parallel corpora: ~53,000 pairs for Guaraní (AmericasNLP 2023 + MultiScript30k), ~7,500 for Bribri (AmericasNLP 2021), and smaller corpora for Nahuatl (Py-Elotl toolkit, Gutierrez-Vasques et al., 2025) and Wixárika (Mager et al., 2018). The cultural taxonomy driving Stage 1 is derived from existing linguistic and ethnographic documentation — no new datasets are created.

Our pipeline has two stages. Stage 1 fine-tunes SmolVLM (2B parameters) via LoRA (rank 16, $\alpha=32$, $lr=2e-4$, up to 10 epochs with early stopping) to produce culturally grounded Spanish descriptions rather than generic ones, using a structured VQA prompting module that interrogates the image before generating ("What ceremony is depicted? What traditional objects are visible?"). Stage 2 retrieves the k most culturally similar translation pairs from a FAISS index built over multilingual sentence embeddings (paraphrase-multilingual-MiniLM-L12-v2) — indexed by cultural concept annotations from Stage 1 rather than surface text — and feeds them as few-shot examples to Gemini 2.5 Flash for translation. Our novel contribution is this coupling: cultural annotations from Stage 1 actively drive retrieval in Stage 2, rather than the stages operating independently. As a stretch goal, we will experiment with fine-tuning NLLB-200 (600M) as an alternative Stage 2 translator, and separately with morphology-aware

post-processing for Nahuatl using FST-based analyzers. All tooling uses HuggingFace transformers and peft for fine-tuning, sentence-transformers and FAISS for retrieval, and pandas/matplotlib for analysis. Experiments run on Google Cloud T4 GPUs (course credits, \$50 budget); translation via Gemini API.

We evaluate primarily using ChrF++, the official task metric, broken down per language and per cultural category to answer RQ2 and RQ3. We supplement with a small human evaluation on ~30 captions using a 3-point cultural accuracy rubric. Visualizations include a per-language ChrF++ bar chart comparing our full pipeline against baselines and two ablations, a language-by-category heatmap, and a retrieval depth ablation curve over $k \in \{3, 5, 8\}$.

Our related work covers eight papers: CultureCLIP (Huang et al., 2025) and CIC (2024) for culturally-aware vision encoding; the AmericasNLP 2026 shared task findings (Bui et al., 2026) and Atlas-Chat (Shang et al., 2025) for low-resource language adaptation; the Gators winning submission (Dhawan et al., 2026) and small LM MT analysis (Song et al., 2026) for our retrieval and translation design; SemEval Task 11 (Muhammad et al., 2025) for per-language/per-category analysis methodology; and the Ojibwe hybrid MT paper (Nguyen et al., 2025) for the morphology stretch goal.

The team of four divides as follows: Romit leads Stage 1 (VLM fine-tuning, cultural VQA module); Nandita owns data and the cultural taxonomy; Mehek builds Stage 2 (retrieval system, LLM prompting, evaluation scripts); Tisha leads analysis and paper writing. Week 1 covers data preparation and access verification; Week 2, an end-to-end generic baseline; Week 3, Stage 1 fine-tuning and taxonomy; Week 4, Stage 2 retrieval experiments and ablations; Week 5, human evaluation, per-category analysis, and visualizations; Week 6, stretch goals and final write-up. The taxonomy must be delivered by the end of Week 2 as it is a dependency for both Stage 1 and Stage 2.